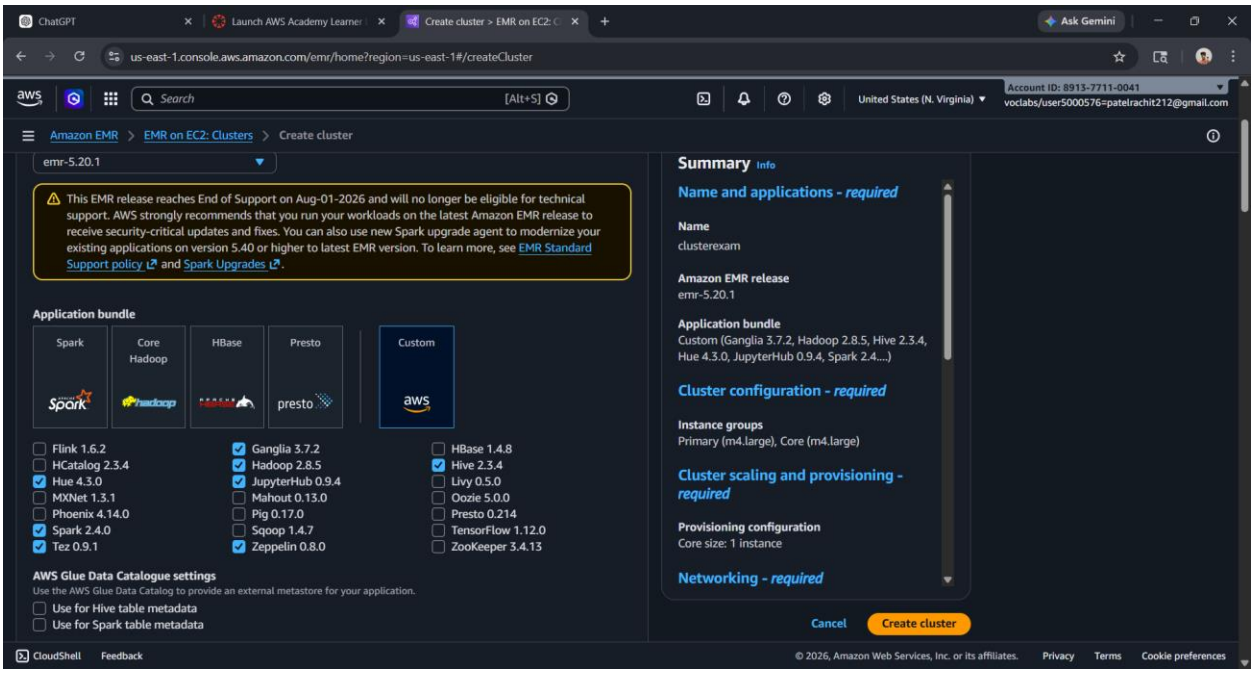
ChatGPT
Launch AWS Academy Learner
Create cluster > EMR on EC2: C
Ask Gemini
us-east-1.console.aws.amazon.com/emr/home?region=us-east-1#/createCluster
Search
[Alt+S]
United States (N. Virginia)
Account ID: 8913-7711-0041
voclabs/user5000576=patelrachit212@gmail.com
Amazon EMR > EMR on EC2: Clusters > Create cluster
emr-5.20.1
This EMR release reaches End of Support on Aug-01-2026 and will no longer be eligible for technical support. AWS strongly recommends that you run your workloads on the latest Amazon EMR release to receive security-critical updates and fixes. You can also use new Spark upgrade agent to modernize your existing applications on version 5.40 or higher to latest EMR version. To learn more, see EMR Standard Support policy and Spark Upgrades.
Application bundle
Spark
Core Hadoop
HBase
Presto
Custom
Flink 1.6.2
HCatalog 2.3.4
Hue 4.3.0
MXNet 1.3.1
Phoenix 4.14.0
Spark 2.4.0
Tez 0.9.1
Ganglia 3.7.2
Hadoop 2.8.5
JupyterHub 0.9.4
Mahout 0.13.0
Pig 0.17.0
Sqoop 1.4.7
Zeppelin 0.8.0
HBase 1.4.8
Hive 2.3.4
Livy 0.5.0
Oozie 5.0.0
Presto 0.214
TensorFlow 1.12.0
ZooKeeper 3.4.13
AWS Glue Data Catalogue settings
Use the AWS Glue Data Catalog to provide an external metastore for your application.
Use for Hive table metadata
Use for Spark table metadata
Summary Info
Name and applications - required
Name
clusterexam
Amazon EMR release
emr-5.20.1
Application bundle
Custom (Ganglia 3.7.2, Hadoop 2.8.5, Hive 2.3.4, Hue 4.3.0, JupyterHub 0.9.4, Spark 2.4....)
Cluster configuration - required
Instance groups
Primary (m4.large), Core (m4.large)
Cluster scaling and provisioning - required
Provisioning configuration
Core size: 1 instance
Networking - required
Cancel
Create cluster
CloudShell
Feedback
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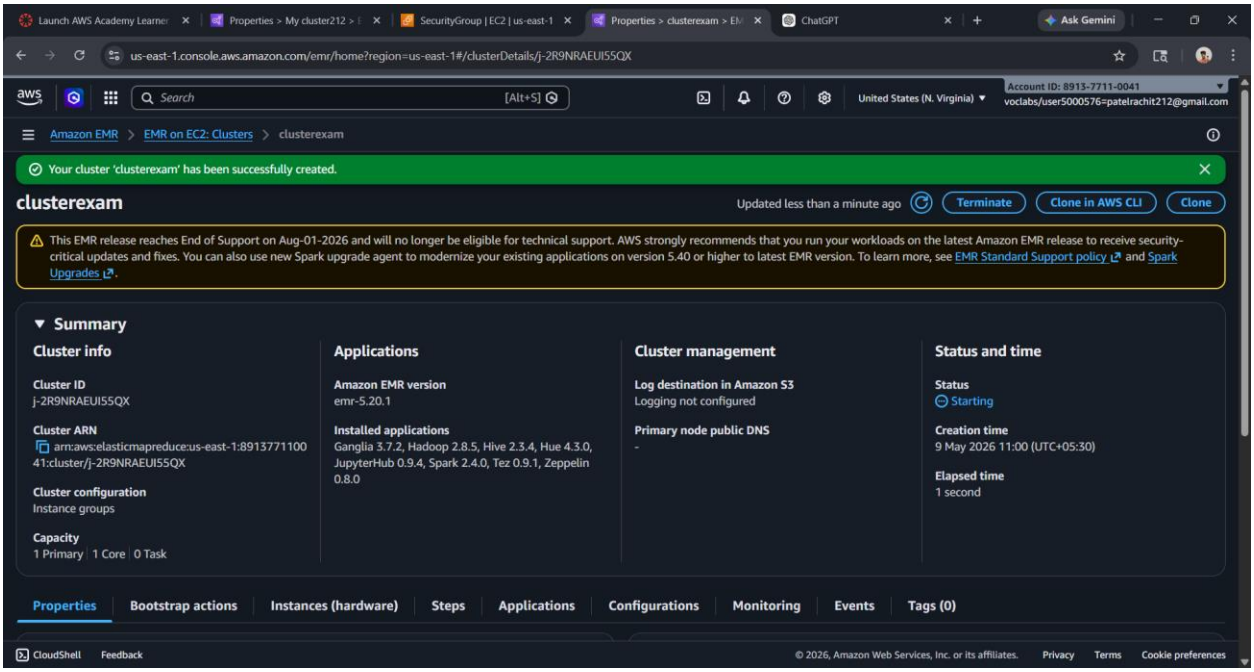
Privacy
Terms
Cookie preferences


Launch AWS Academy Learner
Properties > My cluster212 > I
SecurityGroup | EC2 | us-east-1
Properties > clusterexam > EM
ChatGPT
Ask Gemini
us-east-1.console.aws.amazon.com/emr/home?region=us-east-1#/clusterDetails/j-2R9NRAEUI55QX
Search
[Alt+S]
United States (N. Virginia)
Account ID: 8913-7711-0041
voclabs/user5000576=patelrachit212@gmail.com
Amazon EMR > EMR on EC2: Clusters > clusterexam
Your cluster 'clusterexam' has been successfully created.
clusterexam
Updated less than a minute ago
Terminate
Clone in AWS CLI
Clone
This EMR release reaches End of Support on Aug-01-2026 and will no longer be eligible for technical support. AWS strongly recommends that you run your workloads on the latest Amazon EMR release to receive security-critical updates and fixes. You can also use new Spark upgrade agent to modernize your existing applications on version 5.40 or higher to latest EMR version. To learn more, see EMR Standard Support policy and Spark Upgrades.
Summary
Cluster info
Cluster ID
j-2R9NRAEUI55QX
Cluster ARN
arn:aws:elasticmapreduce:us-east-1:891377110041:cluster/j-2R9NRAEUI55QX
Cluster configuration
Instance groups
Capacity
1 Primary | 1 Core | 0 Task
Applications
Amazon EMR version
emr-5.20.1
Installed applications
Ganglia 3.7.2, Hadoop 2.8.5, Hive 2.3.4, Hue 4.3.0, JupyterHub 0.9.4, Spark 2.4.0, Tez 0.9.1, Zeppelin 0.8.0
Cluster management
Log destination in Amazon S3
Logging not configured
Primary node public DNS
-
Status and time
Status
Starting
Creation time
9 May 2026 11:00 (UTC+05:30)
Elapsed time
1 second
Properties
Bootstrap actions
Instances (hardware)
Steps
Applications
Configurations
Monitoring
Events
Tags (0)
CloudShell
Feedback
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Privacy
Terms
Cookie preferences

ec2-54-82-175-134.compute-1.amazonaws.com (hadoop)

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help X server Exit

Quick connect...

3. ec2-54-82-175-134.compute-1.amaz

/home/hadoop/

Name

..
.aws
.ssh
.bash_profile
.bashrc

Remote monitoring

Follow terminal folder

```
                • MobaXterm Personal Edition v26.3 •
              (SSH client, X server and network tools)

 ➤ SSH session to hadoop@ec2-54-82-175-134.compute-1.amazonaws.com
 |
   • Direct SSH      :  ✔
   • SSH compression :  ✔
   • SSH-browser     :  ✔
   • X11-forwarding  :  ✘  (disabled or not supported by server)

 ➤ For more info, ctrl+click on help or visit our website.

Last login: Sat May  9 06:36:49 2026

       __|  __|_  )
       _|  (     /   Amazon Linux AMI
      ___|\___|___|

https://aws.amazon.com/amazon-linux-ami/2018.03-release-notes/
49 package(s) needed for security, out of 74 available
Run "sudo yum update" to apply all updates.

EEEEEEEEEEEEEEEEEEEE MMMMMMMM           MMMMMMMM RRRRRRRRRRRRRRR
E::::::::::::::::::E M:::::::M         M:::::::M R::::::::::::::R
EE::::::EEEEEEEEE:::E M::::::::M       M::::::::M R:::::RRRRRR:::::R
  E::::E       EEEEE M:::::::::M     M:::::::::M RR::::R      R::::R
  E::::E             M::::::M:::M   M:::M::::::M   R:::R      R::::R
  E::::::EEEEEEEEEE  M:::::M M::M M::M M:::::M   R:::RRRRRR:::::R
  E::::::::::::::::E M:::::M  M:::M:::M  M:::::M   R::::::::::RR
  E::::::EEEEEEEEEE  M:::::M   M:::::M   M:::::M   R:::RRRRRR::::R
  E::::E             M:::::M    M:::M    M:::::M   R:::R      R::::R
  E::::E       EEEEE M:::::M     MMM     M:::::M   R:::R      R::::R
EE::::::EEEEEEEE:::E M:::::M             M:::::M   R:::R      R::::R
E::::::::::::::::::E M:::::M             M:::::M RR::::R      R::::R
EEEEEEEEEEEEEEEEEEEE MMMMMMM             MMMMMMM RRRRRRR      RRRRRR

[hadoop@ip-172-31-48-192 ~]$
```

ip-172-31-48-192 | 5% | 4.88 GB / 7.79 GB | 0.01 Mb/s | 0.01 Mb/s

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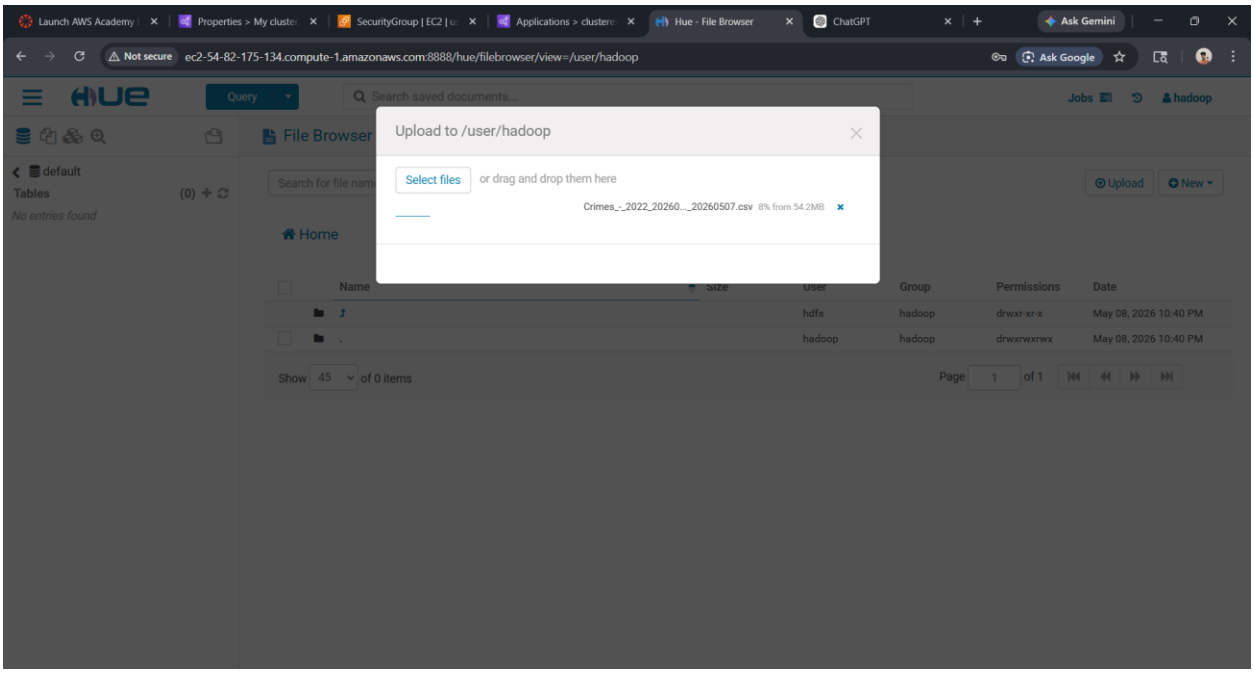
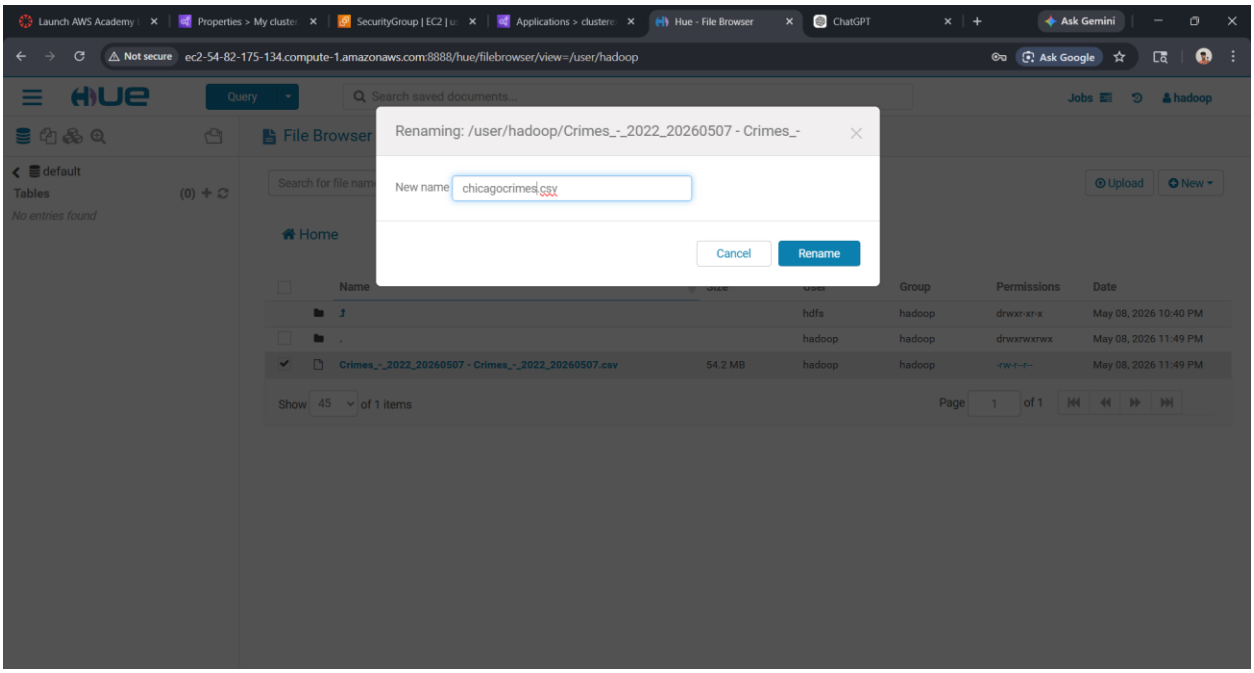
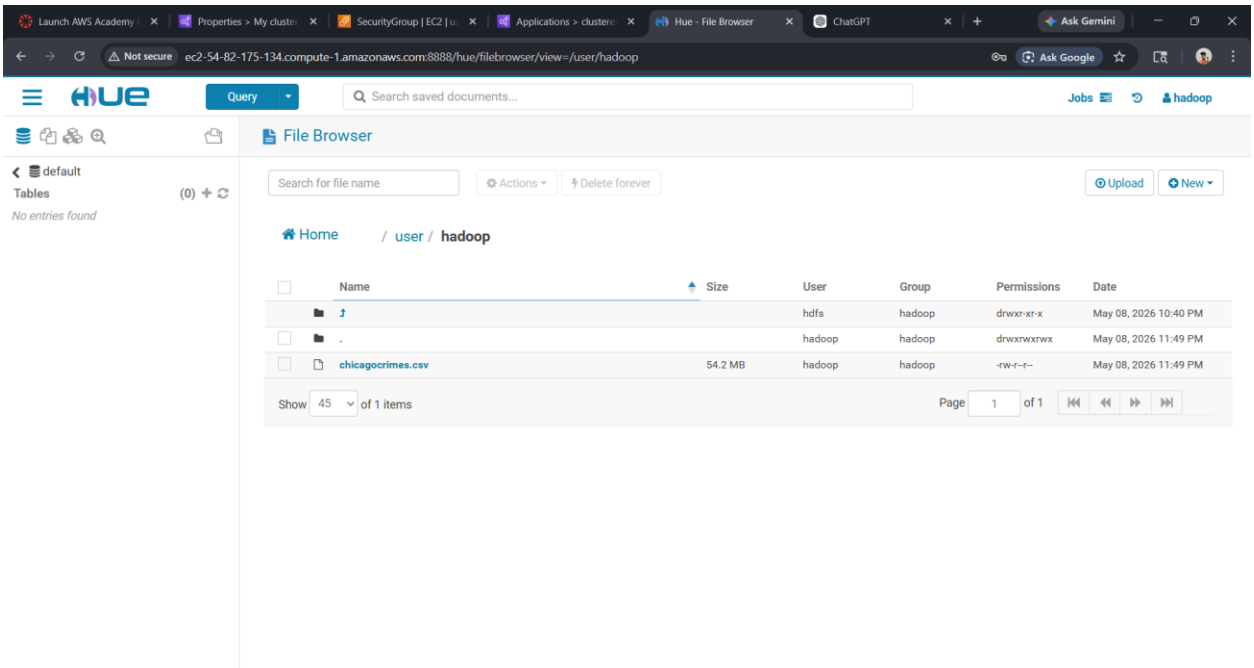

Upload to /user/hadoop
Select files
or drag and drop them here
Crimes_-_2022_20260..._20260507.csv 8% from 54.2MB
File Browser
Home
Upload
New
Name
Size
User
Group
Permissions
Date
hdfs
hadoop
drwxr-xr-x
May 08, 2026 10:40 PM
drwxrwxrwx
Show 45 of 0 items
Page 1 of 1


Renaming: /user/hadoop/Crimes_-_2022_20260507 - Crimes_-
New name
chicagocrimes.csv
Cancel
Rename
File Browser
Home
Upload
New
Name
Size
User
Group
Permissions
Date
hdfs
hadoop
drwxr-xr-x
May 08, 2026 10:40 PM
drwxrwxrwx
May 08, 2026 11:49 PM
Crimes_-_2022_20260507 - Crimes_-_2022_20260507.csv
54.2 MB
-rw-r--r--
Show 45 of 1 items
Page 1 of 1



now using emr hive thorugh mobaxtesm and hue where i have uploaed the csv chicagocrimes.csv in /user/hadoop

part b

table create in with partionitong

```
hive> CREATE DATABASE IF NOT EXISTS crime_db;
OK
Time taken: 1.523 seconds
hive> USE crime_db;
```

c.using hive shell create an external staging table named crimes_raw with the flowing schema:

crime_id INT,

case_number STRing,

crime_date string

block string,

primary_type sting,

description string,

location_desc string

arrest string,

domestic string,

beat int,

district int

ward int,

community_area int,

crime_year int,

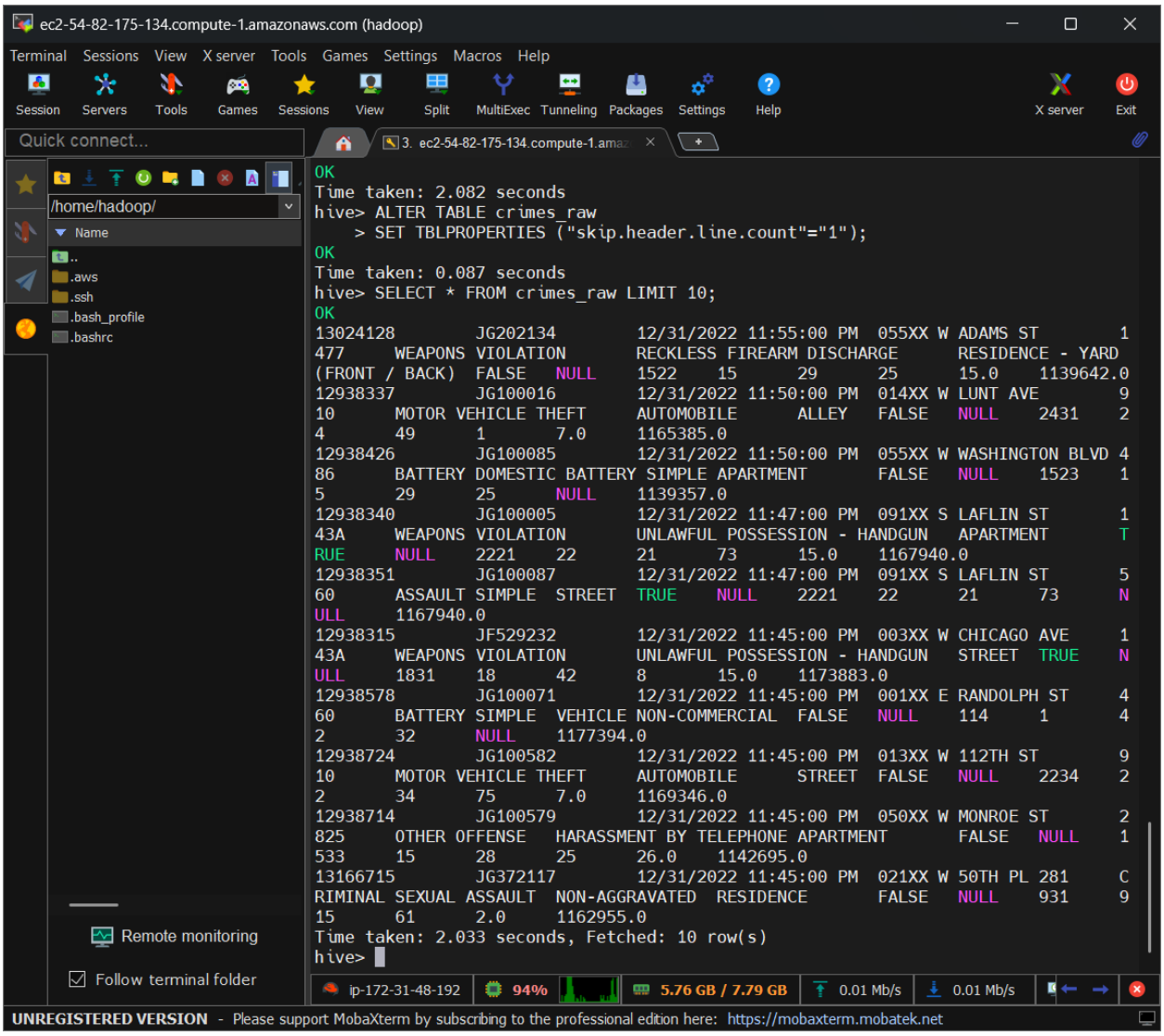
latitude double,

longitude double

```
Time taken: 2.776 seconds
hive> use crime_db;
OK
```

```
hive> CREATE EXTERNAL TABLE crimes_raw (
    >     crime_id INT,
    >     case_number STRING,
    >     crime_date STRING,
    >     block STRING,
    >     primary_type STRING,
    >     description STRING,
    >     location_desc STRING,
    >     arrest STRING,
    >     domestic STRING,
    >     beat INT,
    >     district INT,
    >     ward INT,
    >     community_area INT,
    >     crime_year INT,
    >     latitude DOUBLE,
    >     longitude DOUBLE
    > )
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY ','
    > STORED AS TEXTFILE;
OK
Time taken: 0.16 seconds
```

```
hive> LOAD DATA INPATH '/user/hadoop/chicagocrimes.csv'
    > INTO TABLE crimes_raw;
Loading data to table crime_db.crimes_raw
OK
Time taken: 2.082 seconds
hive>
```



d. create a partitioned and bucketed table named crimes_partitioned , partioned by crime_year and bucketed by district into 4 buckets .load data form crimes_raw into this table

```
hive> SET hive.exec.dynamic.partition=true;
hive> SET hive.exec.dynamic.partition.mode=nonstrict;
hive> SET hive.enforce.bucketing=true;
hive> CREATE TABLE crimes_partitioned (
    >     crime_id INT,
    >     case_number STRING,
    >     crime_date STRING,
    >     block STRING,
    >     primary_type STRING,
    >     description STRING,
    >     location_desc STRING,
    >     arrest STRING,
    >     domestic STRING,
    >     beat INT,
    >     district INT,
    >     ward INT,
    >     community_area INT,
    >     latitude DOUBLE,
    >     longitude DOUBLE
    > )
    > PARTITIONED BY (crime_year INT)
    > CLUSTERED BY (district) INTO 4 BUCKETS
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY ','
    > STORED AS ORC;
OK
Time taken: 0.111 seconds
```

```
hive> SET hive.exec.dynamic.partition=true;
hive> SET hive.exec.dynamic.partition.mode=nonstrict;
hive>
    > SET hive.exec.max.dynamic.partitions=2000;
hive> SET hive.exec.max.dynamic.partitions.pernode=2000;
hive>
    > SET hive.enforce.bucketing=true;
hive> INSERT INTO TABLE crimes_partitioned
    > PARTITION (crime_year)
    > SELECT
    >     crime_id,
    >     case_number,
    >     crime_date,
    >     block,
    >     primary_type,
    >     description,
    >     location_desc,
    >     arrest,
    >     domestic,
    >     beat,
    >     district,
    >     ward,
    >     community_area,
    >     latitude,
    >     longitude,
    >     crime_year
    > FROM crimes_raw;
```

```
    > FROM crimes_raw;
Query ID = hadoop_20260509071545_1e92b2a3-bbac-4c06-b211-c25eda53db0b
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305346
0004)

Map 1: 0/1      Reducer 2: 0/1
Map 1: 0/1      Reducer 2: 0/1
Map 1: 0(+1)/1  Reducer 2: 0/1
Map 1: 0(+1)/1  Reducer 2: 0/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 1/1
Loading data to table crime_db.crimes_partitioned partition (crime_year=null)


         Time taken to load dynamic partitions: 17.565 seconds
         Time taken for adding to write entity : 0.067 seconds
OK
```

Part c

Write and execute on partitioned table

1.count the total number of crime records in the dataset

2.find the top 5 most frequently occurring crime types (primary_type) , ordered by count descending

```
hive> SELECT COUNT(*) AS total_crimes
    > FROM crimes_partitioned;
OK
240065
Time taken: 0.708 seconds, Fetched: 1 row(s)
hive> SELECT primary_type,
    >        COUNT(*) AS crime_count
    > FROM crimes_partitioned
    > GROUP BY primary_type
    > ORDER BY crime_count DESC
    > LIMIT 5;
Query ID = hadoop_20260509071923_3fa917af-0d2d-4b2f-b6d9-642c78b4ab0f
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_177830534641
0004)

Map 1: -/-      Reducer 2: 0/1  Reducer 3: 0/1
Map 1: 0/1      Reducer 2: 0/1  Reducer 3: 0/1
Map 1: 0(+1)/1  Reducer 2: 0/1  Reducer 3: 0/1
Map 1: 0(+1)/1  Reducer 2: 0/1  Reducer 3: 0/1
Map 1: 0(+1)/1  Reducer 2: 0/1  Reducer 3: 0/1
Map 1: 1/1      Reducer 2: 0(+1)/1      Reducer 3: 0/1
Map 1: 1/1      Reducer 2: 1/1  Reducer 3: 0(+1)/1
Map 1: 1/1      Reducer 2: 1/1  Reducer 3: 1/1
OK
810     20114
820     18889
486     18699
910     17160
1320    14238
Time taken: 12.09 seconds, Fetched: 5 row(s)
```

3.calculate the total number of crimes per year (crime_year),ordered by year ascending

ec2-54-82-175-134.compute-1.amazonaws.com (hadoop)

```
1320    14238
Time taken: 12.09 seconds, Fetched: 5 row(s)
hive> SELECT crime_year,
    >        COUNT(*) AS total_crimes
    > FROM crimes_partitioned
    > GROUP BY crime_year
    > ORDER BY crime_year ASC;
Query ID = hadoop_20260509071948_b5b93854-6369-4ce8-9ef7-36455e2b99df
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305346414_
0004)

Map 1: -/-      Reducer 2: 0/1  Reducer 3: 0/1
Map 1: 0/1      Reducer 2: 0/1  Reducer 3: 0/1
Map 1: 0(+1)/1  Reducer 2: 0/1  Reducer 3: 0/1
Map 1: 0(+1)/1  Reducer 2: 0/1  Reducer 3: 0/1
Map 1: 1/1      Reducer 2: 0/1  Reducer 3: 0/1
Map 1: 1/1      Reducer 2: 0(+1)/1      Reducer 3: 0/1
Map 1: 1/1      Reducer 2: 1/1  Reducer 3: 0(+1)/1
Map 1: 1/1      Reducer 2: 1/1  Reducer 3: 1/1
OK
NULL    29
1       4034
2       3946
3       3837
4       2027
5       1393
6       5810
7       3535
8       10566
9       331
10      1277
11      1110
12      671
13      989
14      2241
15      2994
16      2614
17      1633
```

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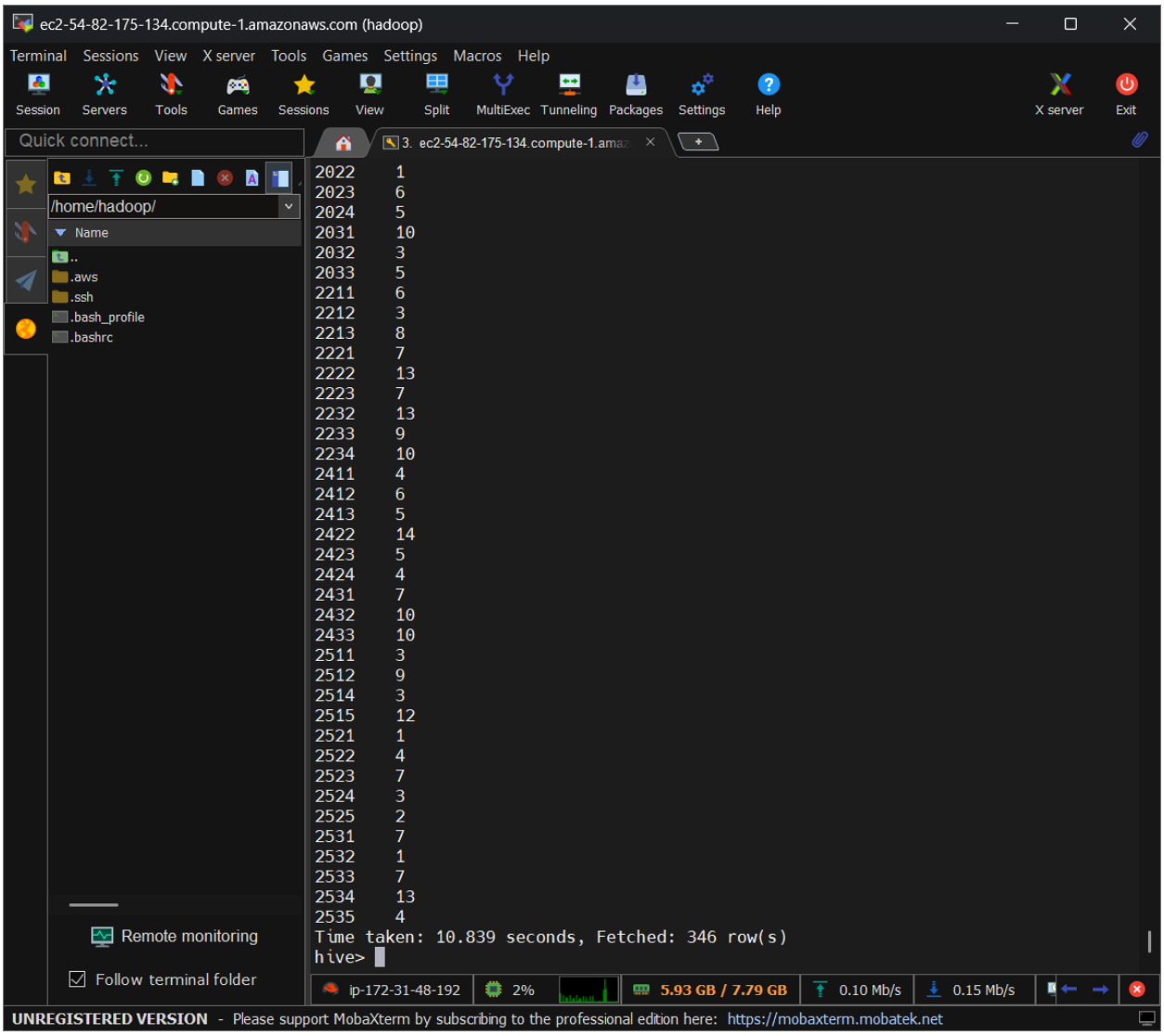

4.find the number of crimes where an arrest was made (arrest ="true')grouped by primary_type.show only crime types with more than 50 arrests.

```
hive> SELECT primary_type,
    >        COUNT(*) AS arrest_count
    > FROM crimes_partitioned
    > WHERE lower(arrest) = 'true'
    > GROUP BY primary_type
    > HAVING COUNT(*) > 50;
Query ID = hadoop_20260509072133_38bdf78f-1fe2-45a0-a2f4-a2ce3dd3a312
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305346414_
0004)

Map 1: -/-      Reducer 2: 0/1
Map 1: 0/1      Reducer 2: 0/1
Map 1: 0(+1)/1  Reducer 2: 0/1
Map 1: 0(+1)/1  Reducer 2: 0/1
Map 1: 0(+1)/1  Reducer 2: 0/1
Map 1: 1/1      Reducer 2: 0(+1)/1
Map 1: 1/1      Reducer 2: 1/1
OK
Time taken: 12.266 seconds
```

5.using a query on the partitioned table , retrieve all crimes for crime_year=2020.demostrate that hive performes partition pruning by noticing the number of partitons scanned in that output

```
hive> SELECT *
    > FROM crimes_partitioned
    > WHERE crime_year = 2020;
OK
```

```
hive> EXPLAIN
    > SELECT *
    > FROM crimes_partitioned
    > WHERE crime_year = 2020;
OK
Plan optimized by CBO.

Stage-0
  Fetch Operator
    limit:-1
    Select Operator [SEL_2]
      Output:["_col0","_col1","_col2","_col3","_col4","_col5","_col6","_col7","_c
ol8","_col9","_col10","_col11","_col12","_col13","_col14","_col15"]
      Filter Operator [FIL_4]
        predicate:(crime_year = 2020)
        TableScan [TS_0]
          Output:["crime_id","case_number","crime_date","block","primary_type","d
escription","location_desc","arrest","domestic","beat","district","ward","communi
ty_area","latitude","longitude"]

Time taken: 0.188 seconds, Fetched: 12 row(s)
hive> [hadoop@ip-172-31-48-192 ~]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.p
roperties Async: false
hive>
```

6. find the top 3 disticts with the highest number of domestic crimes (domestic=”true”).order by count descding